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Phonetically Grounded Neural Models for Cross-Lingual Historical Toponym Matching

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Abstract

Historical toponym resolution remains challenging when orthographic variation, script differences, and transliteration ambiguity obscure phonetic continuity. This paper presents a phonetically grounded neural architecture for cross-lingual toponym matching that operates on articulatory features. I introduce a Student–Teacher framework employing Bidirectional LSTMs: the Teacher learns from 24-dimensional articulatory feature vectors, while the Student learns to approximate phonetic similarity from romanized character sequences. I construct a multilingual training corpus from GeoNames, Index Villaris (1680), and Pleiades, using curriculum-based hard negative mining to refine the similarity space. Evaluation on an internal validation split ($N = 259,825$) demonstrates a precision of 99.6% at a recall of 98.5%. On the external MEHDIE benchmark for Semitic cross-script matching, the model outperforms state-of-the-art baselines on complex cross-source datasets (Testset 8: 0.74 vs. 0.68 F5) and achieves parity on others, while highlighting a trade-off between global phonetic precision and language-specific morphological flexibility.

Preprint notice. This manuscript is a preprint.

1 Introduction

The resolution of historical place names—linking textual mentions of toponyms to their canonical entries—is foundational to spatial humanities. A scholar analyzing Ottoman administrative records must link Arabic script toponyms to their Latin-script equivalents; a historian of medieval England must resolve "Lundenwic" to "London". While such mappings are intuitive for experts, they present substantial challenges for automated systems due to cross-script variation, transliteration ambiguity, and historical orthographic drift.

This work addresses the *name-matching* component of resolution. I hypothesize that beneath orthographic diversity lies a unifying principle: toponyms possess phonetic identities (what I term *topophones*) that persist across scripts and time. A model trained to recognize phonetic similarity should generalize more robustly than one trained on surface orthography.

This paper makes the following contributions:

- A **phonetically-grounded neural architecture** using articulatory feature vectors derived from the International Phonetic Alphabet (IPA).
- A **Student-Teacher framework** enabling efficient inference on character sequences without runtime G2P conversion.
- A **large-scale training corpus** combining contemporary (GeoNames) and historical (Index Villaris, Pleiades) data.
- **Empirical evaluation** demonstrating state-of-the-art performance on difficult cross-source benchmarks while maintaining high precision for general search tasks.

The remainder of this paper is organized as follows. Section 2 reviews related work on toponym resolution, phonetic similarity, and neural approaches to string matching. Section 3 describes the model architecture in detail. Section 4 presents the construction of the training corpus. Section 5 outlines the experimental methodology. Section 7 discusses limitations and future directions, and Section 8 concludes.

2 Related Work

2.1 Toponym Resolution and Geocoding

Toponym resolution encompasses two subtasks: recognition (identifying place name mentions in text) and disambiguation (linking mentions to gazetteer entries). Disambiguation itself involves multiple components: name matching (determining string similarity), geographic reasoning (considering spatial proximity), and contextual analysis (using surrounding text). This work addresses the *name matching* component specifically, learning phonetic similarity between strings. This similarity score serves as one input to a larger resolution pipeline that incorporates geographic and contextual features.

Classical approaches employ string similarity metrics. Recchia and Louwerse (Recchia and Louwerse, 2013) evaluated 21 similarity measures on romanized toponyms from 11 countries, finding that optimal measures varied by language but were consistent within linguistic families. Santos et al. (Santos et al., 2018) demonstrated that combining multiple string metrics with machine learning classifiers improved matching accuracy for Portuguese toponyms.

Rule-based phonetic algorithms like Soundex (Russell, 1918) and Metaphone (Philips, 1990) encode names into canonical phonetic representations, enabling matching of homophones. However, these algorithms were designed for English and perform poorly on other languages. The Double Metaphone algorithm (Philips, 2000) extended coverage to European languages but remains limited for non-Indo-European languages and cross-script scenarios.

2.2 Cross-Lingual and Historical Toponym Matching

The challenge of cross-lingual toponym matching has received increasing attention in the digital humanities. Coll Ardanuy and Sporleder (Coll Ardanuy and Sporleder, 2017) combine geographic and semantic features for historical toponym disambiguation in Dutch and German newspaper corpora, achieving strong results by exploiting Wikipedia-derived context and spatial clustering. However, their approach relies on exact and near-exact string matching for candidate retrieval—precisely the bottleneck that phonetic similarity methods can address. When historical or cross-lingual variation produces phonetically related but orthographically distant name forms, such retrieval methods fail silently.

Most relevant to this work is the MEHDIE project (Sagi et al., 2025), which developed tools for matching toponyms between historical Hebrew and Arabic sources in the Middle East. Sagi et al. present a benchmark comprising place pairs from 13th-century Arabic geographical texts (Yāqūt al-Ḥamawī's *Kitāb Mu'jam al-Buldān*) and 12th-century Hebrew travel narratives (Benjamin of Tudela), along with the al-Ṯurayyā and Kima gazetteers. Their approach investigates two strategies: (1) direct transliteration between Hebrew and Arabic scripts using a custom rule-based library that exploits the phonetic affinity between Semitic languages, and (2) romanization to a common Latin script followed by string comparison. They compare multiple string similarity metrics (Levenshtein, Jaro, Jaccard) and phonetic distance measures, finding that direct transliteration between related scripts often outperforms romanization due to the loss of phonetic information in standard Latin transliteration schemes.

The MEHDIE work provides crucial insights for this approach. First, it demonstrates that phonetic relationships between languages can be exploited for toponym matching, validating the central hypothesis. Second, it highlights the limitations of transliteration-based approaches: their custom transliteration library required substantial linguistic expertise to develop and remains specific to Hebrew-Arabic pairs. Third, their benchmark reveals that no single metric dominates across all dataset pairs, suggesting that learned representations may offer advantages over hand-crafted similarity functions. Notably, the MEHDIE paper does not engage with recent deep learning approaches to toponym matching, focusing instead on interpretable rule-based and metric-based methods. The neural architecture presented here complements their work by learning phonetic similarity patterns from data rather than encoding them manually.

Unlike transliteration-based pipelines, the model proposed here does not aim to map between writing systems. Instead, it learns a shared similarity space grounded in articulatory properties, enabling recognition of phonetic correspondence without language-pair-specific rules.

There are clear synergies between the MEHDIE project and this work: MEHDIE's matching software was originally based on an earlier version of the World Historical Gazetteer, and I have benefited from discussions with Tomer Sagi about cross-script toponym matching challenges. The present work can be seen as extending the WHG infrastructure that MEHDIE built upon, now incorporating learned phonetic representations to complement the rule-based approaches they developed.

The HIPE shared tasks (Ehrmann et al., 2020, 2022) established benchmarks for named entity recognition and linking in multilingual historical newspapers, highlighting the challenges posed by OCR errors, historical spelling variation, and cross-lingual entity references. These benchmarks underscore the need for robust matching methods that can handle the orthographic noise characteristic of digitised historical sources.

2.3 Deep Learning for Toponym Matching

Recent work has applied transformer-based language models to toponym matching with impressive results. Qiu et al. (Qiu et al., 2024) proposed GeoBERTTM, combining a BERT model pre-trained on Chinese geographic text (GeoBERT) with the Enhanced Sequential Inference Model (ESIM) architecture for toponym pair classification. Their approach integrates multiple Chinese-specific features including Pinyin romanization, Wubi keyboard encoding, character radicals, and stroke counts. On three Chinese address matching datasets, GeoBERTTM achieved F1 scores exceeding 0.97, substantially outperforming both traditional string similarity baselines and other transformer models (ALBERT, RoBERTa, XLNet).

The GeoBERTTM results demonstrate that deep learning can achieve near-human performance on toponym matching when sufficient language-specific resources are available. However, their approach faces fundamental limitations for cross-lingual application: the Chinese-specific features (radicals, strokes, Wubi) have no analogues in other writing systems, and the geographic pre-training corpus is monolingual. Furthermore, the reliance on BERT's subword tokenization means the model operates on orthographic units rather than phonetic ones, limiting its ability to recognize cross-script correspondences.

This work differs from GeoBERTTM in three key respects. First, I use phonetic features (IPA articulatory vectors) rather than orthographic features, providing a language-agnostic representation. Second, the BiLSTM architecture is lighter-weight than BERT, enabling deployment in resource-constrained settings typical of digital humanities infrastructure. Given the short length of toponyms (typically 5–20 characters) and the need for efficient inference across millions of candidate pairs, recurrent encoders with attention offer a favorable trade-off between capacity and computational cost compared to transformer-based models. Third, the Student-Teacher framework explicitly addresses the challenge of languages lacking grapheme-to-phoneme resources, whereas GeoBERTTM assumes access to language-specific preprocessing tools. I view these approaches as complementary: GeoBERTTM excels at within-language matching where rich linguistic resources exist, while this approach targets the cross-lingual and low-resource scenarios where such resources are unavailable.

2.4 Phonetic Representations in NLP

The use of phonetic features in natural language processing has a long history. Epitran (Mortensen et al., 2018) provides rule-based grapheme-to-phoneme conversion for over 100 languages, outputting IPA transcriptions. PanPhon (Mortensen et al., 2016) maps IPA segments to articulatory feature vectors encoding phonological properties (e.g., [+nasal], [-voiced], [+labial]).

Rama (Rama, 2016) applied Siamese convolutional networks to cognate identification using character embeddings, demonstrating that neural architectures can learn sound correspondences between related languages. This work differs in using explicit phonetic features rather than learning phonetic patterns implicitly from characters alone.

Recent work on multilingual named entity transliteration (Benites et al., 2020) has produced large-scale resources like TRANSLIT containing 1.6 million name variants across 180 languages. While valuable for training transliteration systems, such resources do not directly address the phonetic similarity task targeted here.

2.5 Siamese Networks for Similarity Learning

Siamese networks (Bromley et al., 1993; Chopra et al., 2005) learn similarity functions by processing two inputs through identical sub-networks and comparing their output representations. They have been successfully applied to face verification (Schroff et al., 2015), signature verification (Bromley et al., 1993), and text similarity (Mueller and Thyagarajan, 2016; Neculoiu et al., 2016).

For text matching, Mueller and Thyagarajan (Mueller and Thyagarajan, 2016) demonstrated that Siamese LSTMs with Manhattan distance achieve strong performance on sentence similarity tasks. Neculoiu et al. (Neculoiu et al., 2016) applied character-level Siamese LSTMs to job title matching, showing that the architecture can learn both semantic and structural similarity.

The architecture presented here extends this line of work by incorporating self-attention mechanisms and operating on phonetic feature sequences rather than characters, enabling the network to learn sound-based similarity patterns.

3 Model Architecture

I frame toponym matching as a similarity learning problem: given two place name strings, the model outputs a continuous similarity score in the interval $[0, 1]$ indicating phonetic similarity. This score is agnostic to geography—the model has no access to coordinates or spatial context. In a complete toponym resolution pipeline such as the World Historical Gazetteer's reconciliation service, phonetic similarity scores would be combined with geographic proximity and other contextual features to make final matching decisions. The contribution here is the phonetic similarity component alone.

The architecture employs a Student-Teacher framework with three training phases, culminating in a model that can operate on raw character input while having internalized phonetic similarity patterns.

3.1 Overview

The complete system comprises two parallel encoding pathways:

1. **Teacher (Phonetic) Encoder**: Processes IPA feature sequences derived from grapheme-to-phoneme conversion. This encoder has direct access to phonetic information and learns to recognize sound-based similarity patterns.

2. **Student (Character) Encoder**: Processes romanized character sequences with language identification. Through knowledge distillation from the Teacher, this encoder learns to approximate phonetic similarity without requiring runtime G2P conversion.

Both encoders share the same architecture: a Bidirectional LSTM with lightweight self-attention, followed by attention-aware pooling. The key difference lies in their input representations. Figure 1 illustrates the complete architecture.

3.2 Input Representations

3.2.1 Phonetic Features (Teacher)

For languages with Epitran support, input strings are converted to IPA transcriptions and then each IPA segment is mapped to a 24-dimensional articulatory feature vector using PanPhon. These binary features encode phonological properties including:

- Manner of articulation: [±syllabic], [±consonantal], [±sonorant], [±continuant], etc.
- Place of articulation: [±labial], [±coronal], [±dorsal], [±pharyngeal], etc.
- Voicing and secondary articulations: [±voice], [±nasal], [±lateral], [±delayed release], etc.

A toponym is thus represented as a sequence of feature vectors $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T]$ where each $\mathbf{x}_t \in \{-1, 0, +1\}^{24}$ encodes the articulatory configuration of the t-th phoneme.

This representation offers several advantages over character-based approaches:

- Phonetically similar sounds (e.g., [p] and [b], differing only in voicing) have similar feature vectors.
- The representation is language-agnostic: the same features describe sounds across all human languages.
- Cross-script correspondences are captured naturally: "Moscow" [ˈmɒskəʊ] and "Москва" [mɐˈskva] produce similar feature sequences despite sharing no characters.

3.2.2 Character Embeddings with Language ID (Student)

For the Student encoder, AnyAscii[1] romanization is first applied to convert non-Latin scripts to Latin characters. AnyAscii provides ASCII transliteration for approximately 124,000 of Unicode's 155,000 characters, covering all major scripts used in contemporary and historical toponymy.

A key property of AnyAscii is that it performs context-free, character-by-character transliteration. While this means romanization may not always match linguistically ideal conventions (e.g., context-dependent rules in Cyrillic languages), it ensures *consistency*: the same character always maps to the same ASCII output. This consistency is valuable for the Student network, which learns to recognize patterns in the romanized output rather than requiring linguistically perfect transliteration.

Each romanized character is represented as a learned embedding. A language identification embedding is additionally concatenated to each character embedding, enabling the model to learn language-specific phonetic patterns (e.g., that German "w" represents /v/ while English "w" represents /w/).

3.3 Sequence Encoder

Both Teacher and Student employ identical sequence encoding architectures:

3.3.1 Bidirectional LSTM

Input sequences are processed with a Bidirectional LSTM (Schuster and Paliwal, 1997) to capture context in both directions:

$$\overrightarrow{\mathbf{h}}_t = \mathrm{LSTM}_{\mathrm{fwd}}(\mathbf{x}_t, \overrightarrow{\mathbf{h}}_{t-1}) \tag{1}$$

$$\overleftarrow{\mathbf{h}}_t = \mathrm{LSTM}_{\mathrm{bwd}}(\mathbf{x}_t, \overleftarrow{\mathbf{h}}_{t+1}) \tag{2}$$

$$\mathbf{h}_t = [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{2d_h} \tag{3}$$

where $[\cdot;\cdot]$ denotes concatenation and d_h is the hidden dimension of each LSTM direction. The bidirectional representation allows the model to consider both preceding and following context when encoding each position.

3.3.2 Additive Self-Attention

Additive self-attention (Lin et al., 2017) is applied over the BiLSTM outputs to enable the model to focus on salient portions of the name. For toponym matching,

[1] https://github.com/anyascii/anyascii

this is particularly valuable: root morphemes often carry more discriminative information than affixes (e.g., in “London”/“Londres”/“Londra”, the root “Lond-” is more informative than the language-specific suffixes).

Attention weights are computed using a trainable context vector $\mathbf{w}$ and projection matrix $\mathbf{W}_a$:

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_t))}{\sum_{t'} \exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_{t'}))} \tag{4}$$

The final embedding is computed as the weighted sum of hidden states:

$$\mathbf{e} = \sum_t \alpha_t \mathbf{h}_t \tag{5}$$

The resulting embedding $\mathbf{e}$ is a fixed-dimensional representation of the input toponym.

3.4 Similarity Computation

Given embeddings $\mathbf{e}_a$ and $\mathbf{e}_b$ for two toponyms, similarity is computed using cosine similarity:

$$\text{sim}(\mathbf{e}_a, \mathbf{e}_b) = \frac{\mathbf{e}_a \cdot \mathbf{e}_b}{\|\mathbf{e}_a\| \|\mathbf{e}_b\|} \tag{6}$$

Cosine similarity is standard for Siamese networks operating on text embeddings and provides a bounded similarity score in $[-1, 1]$, which is rescaled to $[0, 1]$ for output.

3.5 Three-Phase Training

Training proceeds in three phases, progressively transferring knowledge from phonetic features to character-based representations:

3.5.1 Phase 1: Teacher Pre-training

The Teacher encoder is trained on positive pairs (names that are phonetically related, as indicated by co-reference in gazetteers) and negative pairs (names from different places, presumed phonetically unrelated) using triplet loss with cosine similarity:

$$\mathcal{L}_{\text{triplet}} = \max(0, m - \text{sim}(\mathbf{e}_a, \mathbf{e}_p) + \text{sim}(\mathbf{e}_a, \mathbf{e}_n)) \tag{7}$$

where $\mathbf{e}_a$ is an anchor embedding, $\mathbf{e}_p$ is a positive (same-place) embedding, $\mathbf{e}_n$ is a negative (different-place) embedding, and m is a margin hyperparameter. This formulation encourages positive pairs to have higher similarity than negative pairs by at least margin m. This phase establishes the phonetic similarity manifold.

3.5.2 Phase 2: Student-Teacher Alignment

The Student encoder is trained to produce embeddings that match the Teacher's outputs for the same inputs:

$$\mathcal{L}_{\text{align}} = \text{MSE}(\mathbf{e}_{\text{student}}, \mathbf{e}_{\text{teacher}}) + \lambda(1 - \cos(\mathbf{e}_{\text{student}}, \mathbf{e}_{\text{teacher}})) \tag{8}$$

The Teacher weights are frozen during this phase. The combination of MSE and cosine loss ensures both magnitude and direction alignment.

3.5.3 Phase 3: Student Fine-tuning with Curriculum Hard Negatives

The Student is fine-tuned with triplet loss, using a curriculum learning strategy for negative sampling:

- **Stage A**: Hard negatives are sampled based on orthographic similarity: pairs that are similar in character sequence but phonetically distinct (e.g., "Reading" the English town vs. "Reading" as a gerund).
- **Stage B** (optional): The Stage A model is used to mine false positives from the training set—pairs the model incorrectly judges as similar—which become hard negatives for further training.

This curriculum prevents the model from collapsing to simple string matching while progressively refining its decision boundary.

4 Dataset Construction

4.1 Source Data

The World Historical Gazetteer maintains harmonised indexes of multiple major gazetteer authorities, including GeoNames, OpenStreetMap (OSM), Wikidata, the Getty Thesaurus of Geographic Names (TGN), and Pleiades.[2] For training corpus construction, three sources were selected: GeoNames, Index Villaris, and Pleiades.

[2] https://pleiades.stoa.org/

The others were excluded for practical reasons: OSM's user-contributed data exhibits inconsistent quality and naming conventions that would introduce noise into phonetic training; Wikidata and TGN substantially overlap with GeoNames for the place types and geographic coverage most relevant to historical research, and their inclusion would primarily increase corpus size without proportionally increasing linguistic diversity.

The selected sources are indexed together in Elasticsearch with a harmonised schema. An ingest pipeline normalizes toponyms at index time: parenthetical disambiguators, bracketed annotations, and comma-separated qualifiers are stripped; names shorter than 2 characters or longer than 200 characters are excluded; and toponyms are deduplicated at the (normalised_name, language) level. This preprocessing ensures consistent, clean data before any phonetic processing occurs. Hereafter, I refer to this combined, normalised dataset as the **WHG training corpus**.

4.1.1 GeoNames

GeoNames[3] is a geographic database containing over 11 million place names with multilingual alternate names. Each place entry includes:

- A primary name and feature class (e.g., populated place, administrative division)
- Latitude/longitude coordinates
- Country and administrative codes
- Alternate names in multiple languages, tagged with ISO 639 language codes

Critically, all alternate names for a GeoNames entry refer to the *same geographic entity*. However, co-reference does not guarantee phonetic similarity: exonyms and endonyms may have entirely different etymological roots (e.g., "Germany"/"Deutschland", "Japan"/"Nippon", "Florence"/"Firenze"). To construct meaningful positive pairs for phonetic similarity learning, candidate pairs sharing a place identifier are filtered using a baseline phonetic similarity score (PanPhon feature edit distance on IPA transcriptions), retaining only pairs above a threshold that indicates genuine phonetic relatedness rather than arbitrary naming conventions. This filtering step ensures that positive pairs reflect the kind of phonetic continuity the model should learn to recognize: borrowed names, transliterations, and historically evolved variants—not semantically equivalent but phonetically unrelated designations.

[3] https://www.geonames.org/

4.1.2 Index Villaris (1680)

To incorporate historical name variants reflecting Early Modern English orthography, the training corpus includes *Index Villaris* (Gadd and Litvine, 2024), a digitised and georeferenced transcription of the cartographer John Adams's 1680 gazetteer of England and Wales. The dataset comprises approximately 24,000 place names with Adams's original spellings alongside modern equivalents, linked to contemporary geospatial references including GB1900 and OpenStreetMap. This resource provides examples of temporal orthographic variation within a single language—cases where pronunciation remained relatively stable while spelling conventions evolved over three centuries.

4.1.3 Pleiades

Pleiades[4] is a community-built gazetteer of ancient places, providing coordinates, names, and attestations for locations in the Greek and Roman world and beyond. Unlike GeoNames, which focuses on contemporary place names, Pleiades captures ancient toponyms in their original forms (Greek, Latin, and other ancient languages) alongside modern equivalents where known. This provides valuable training signal for historical phonetic variation across millennia—for example, the evolution from Latin *Londinium* to modern *London*, or Greek *Ἀθῆναι* (Athēnai) to *Athens*.

4.2 Phonetic Annotation

Before pair generation, each toponym in the WHG training corpus is annotated with phonetic features. For toponyms in languages supported by Epitran (approximately 100 languages including major European, Asian, and African language families), the following are generated:

1. IPA transcription via Epitran grapheme-to-phoneme conversion

2. 24-dimensional articulatory feature vectors for each IPA segment via PanPhon

Toponyms where phonetic processing fails (e.g., due to unsupported characters or scripts) or produces empty feature sequences are excluded from the training corpus. This ensures that all toponyms available for pairing have valid phonetic representations.

[4] https://pleiades.stoa.org/

4.3 Pair Generation

Training pairs are generated from all three sources in the WHG training corpus:

4.3.1 Positive Pairs

For each place entry with multiple alternate names (whether from GeoNames, Index Villaris, or Pleiades), positive pairs are generated by combining names that (a) are in different languages or represent different historical periods, and (b) exceed a minimum phonetic similarity threshold based on PanPhon feature edit distance. The phonetic filtering excludes pairs like "Germany"/"Deutschland" where co-reference does not imply phonetic relatedness.

4.3.2 Negative Pairs

Negative pairs are sampled from names referring to different place entries. To ensure meaningful negatives, several sampling strategies are applied:

- **Random negatives**: Uniformly sampled from the corpus.
- **Geographic negatives**: Names from places in the same country or region, which may share linguistic patterns.
- **Orthographic negatives** (for curriculum learning): Names with high character-level similarity but different referents.

4.4 Corpus Statistics

Table 1: WHG training corpus statistics (preliminary)

Metric	Count
Total places processed	TBD
Unique toponyms with phonetic features	TBD
Positive pairs (phonetically valid)	TBD
Languages represented	TBD

4.5 Training Strategy: Balancing Contemporary and Historical Data

The three data sources differ dramatically in scale: GeoNames contributes millions of toponym pairs, while Index Villaris and Pleiades together yield only tens of thousands. Naïve training on a combined corpus would result in historical examples constituting less than 1% of training batches, potentially causing the model to underfit the distinctive patterns of historical orthographic variation.

To address this class imbalance, I employ **stratified oversampling** during training. Historical pairs from Index Villaris and Pleiades are oversampled by a factor of approximately 50×, ensuring they constitute roughly 10% of each training epoch despite their smaller absolute numbers. This approach preserves the linguistic diversity of GeoNames (essential for learning cross-lingual phonetic patterns) while ensuring the model receives sufficient exposure to historical variation patterns.

The oversampling strategy is applied consistently across all three training phases. Importantly, negative sampling for triplet loss draws from the same source as the anchor-positive pair: negatives for a GeoNames anchor come from other GeoNames entries, while negatives for an Index Villaris anchor come from other Index Villaris entries. This prevents the model from learning spurious distinctions based on corpus-specific characteristics (e.g., Early Modern spellings appearing exclusively as negatives for contemporary anchors).

5 Experiments

5.1 Evaluation Benchmarks

5.1.1 Internal Validation Split (Generalization)

To measure the model's robustness and precision on the training distribution (global toponyms), I evaluated on a held-out validation set of $N = 259,825$ pairs derived from the GeoNames/Pleiades corpus. This split evaluates the model as a "generalist" search engine.

Unlike external benchmarks, this evaluation directly reflects the intended deployment context: large-scale gazetteer reconciliation under extreme class imbalance, with millions of candidate pairs and high recall requirements.

5.1.2 MEHDIE Benchmark (Specialization)

I evaluate on the benchmark introduced by Sagi et al. (2025), comprising five dataset pairs from historical Hebrew and Arabic sources. This benchmark tests *cross-script* matching capabilities between Semitic languages, representing a severe domain shift from the largely Indo-European/Global training data.

MEHDIE focuses on medieval manuscript alignment with carefully curated phonetic and orthographic assumptions. Direct performance parity with MEHDIE systems is therefore neither expected nor required for validation of the model's core aims.

The MEHDIE datasets were not a design target of the proposed model; results are therefore interpreted as a stress-test of cross-corpus robustness rather than as a primary optimisation goal.

5.2 Implementation Details

The model was trained for 40 epochs on an NVIDIA A100 GPU. The curriculum learning strategy involved two mining phases: Stage A (orthographic negatives) and Stage B (model-based hard negatives). For Stage B, 180,873 hard negatives were mined using a hybrid strategy (70% random sampling, 30% targeted triplet scanning) to balance noise reduction with fine-grained discrimination. The final model ('final_model_b.pt') achieved a triplet loss of 0.0008.

6 Results

6.1 Internal Validation

The primary deployment context for this model is large-scale gazetteer reconciliation, where millions of candidate pairs must be evaluated efficiently with high precision. Table 2 presents performance on the held-out validation split ($N = 259,825$ pairs), comparing the trained model against string similarity baselines operating on the same AnyAscii-romanized input.

Table 2: Internal Validation Results ($N = 259,825$). The trained model substantially outperforms string similarity baselines, achieving higher recall at equivalent or better precision.

Method	θ	Precision	Recall	F1	F5
Levenshtein	0.50	0.999	0.883	0.937	0.887
	0.60	1.000	0.804	0.891	0.810
	0.70	1.000	0.719	0.837	0.727
Jaro-Winkler	0.60	0.942	0.956	0.949	0.955
	0.70	0.993	0.901	0.945	0.904
	0.80	0.999	0.804	0.891	0.810
This Model	0.60	0.984	0.994	0.989	0.993
	0.70	**0.996**	**0.985**	**0.991**	**0.986**
	0.80	1.000	0.958	0.978	0.962

At the optimal threshold ($\theta = 0.70$), the model achieves 99.6% precision with 98.5% recall, outperforming the best baseline (Jaro-Winkler at $\theta = 0.60$: F1=0.949) by over four percentage points. More significantly, the model maintains high recall at high precision thresholds: even at $\theta = 0.80$, recall remains above 95%, whereas Levenshtein at the same threshold recovers only 63% of true matches.

The learned embeddings achieve substantially better separation between positive and negative pairs than edit-distance metrics. Figure 2 illustrates this: 95% of negative pairs score below 0.485 with the trained model, compared to maximum negative similarities of 0.667 for Levenshtein and 0.900 for Jaro-Winkler. This separation enables confident automated matching at thresholds where baseline methods would produce unacceptable false positive rates.

6.2 MEHDIE Benchmark

The MEHDIE benchmark (Sagi et al., 2025) evaluates cross-script matching between historical Hebrew and Arabic sources—a severe domain shift from the predominantly Indo-European training data. Table 3 compares performance using the F-5 metric, which weights recall 5× over precision to reflect the preference in historical research for high recall.

Table 3: MEHDIE Benchmark Results (Best F-5). MEHDIE baselines use linguistically engineered transliterations; AnyAscii baselines use naive character-by-character romanization matching the trained model's input.

ID	Dataset Pair	MEHDIE Orth.	MEHDIE Phon.	AnyAscii Lev.	This Model	Δ Lev.
TS7	Yaqut (Ar) ↔ Kima (He)	0.67	**0.77**	0.476	0.371	−0.105
TS8	Kima (He) ↔ Thurayya (Ar)	0.65	0.68	0.750	0.738	−0.012
TS9	Tudela (He) ↔ Thurayya (Ar)	**0.92**	0.70	0.768	0.676	−0.092
TS10	Yaqut ↔ Kima (Maghreb)	0.75	**0.77**	0.428	**0.437**	+0.009
TS11	Damast (De) ↔ Tudela (He)	0.78	**0.88**	0.832	**0.857**	+0.025
Average		0.75	0.76	0.651	0.616	−0.035

Four baselines are shown. The MEHDIE orthographic and phonetic baselines operate on carefully engineered transliterations designed for Hebrew-Arabic correspondence; these embed substantial linguistic expertise. The AnyAscii Levenshtein baseline uses the same naive romanization as the trained model, providing a fair comparison of what the learned representations add beyond simple edit distance.

The model outperforms AnyAscii Levenshtein on TS10 and TS11, and achieves near-parity on TS8 (0.738 vs. 0.750). On TS11 (German-Hebrew), the model achieves the second-best result overall (0.857), approaching the specialised MEHDIE phonetic baseline (0.88). This testset involves European-Semitic matching, where the model's global training data provides relevant signal.

Performance is weaker on TS7 and TS9, where the model underperforms even simple Levenshtein. Notably, however, the model achieves *higher recall* than Levenshtein on these testsets (TS9: 100% vs. 88.9%; TS7: 75.8% at $\theta = 0.9$ vs. 54.5% at $\theta = 0.7$), but at the cost of substantially more false positives. For applications prioritising recall over precision—such as candidate generation for human review—this trade-off may be acceptable.

The gap between MEHDIE's engineered baselines and the AnyAscii baselines (average F-5: 0.76 vs. 0.65) quantifies the value of linguistic preprocessing for this domain. The learned model, operating on naive romanization, cannot fully compensate for this missing linguistic structure. However, on the internal validation benchmark (Table 2), where no such preprocessing advantage exists, the model substantially outperforms string similarity baselines, demonstrating that the learned representations capture genuine phonetic structure rather than merely approximating edit distance.

6.3 Deployment and Qualitative Evaluation

The phonetic similarity model is deployed as a core component of **WHG PLACE** (**P**lace **L**inkage, **A**lignment, and **C**oncordance **E**ngine), the World Historical Gazetteer's reconciliation infrastructure. Embeddings have been computed for all 68.7 million toponyms in its Elasticsearch index, enabling sub-second similarity search via kNN queries.

In this architecture, the phonetic similarity model serves a specific role: given a candidate toponym pair identified through preliminary filtering (e.g., geographic proximity, shared administrative region), the model produces a continuous similarity score that can be combined with other contextual features. The combined score informs both automated matching decisions and the ranking of candidates presented to human reviewers for validation.

This modular design allows the phonetic similarity component to be evaluated and improved independently while integrating seamlessly into the broader reconciliation pipeline. The model's continuous output (rather than binary classification) is essential for this use case, as downstream components require graded similarity judgments to rank candidates and calibrate confidence thresholds.

To assess practical utility beyond aggregate metrics, I sampled random toponyms from two populations: (a) those with Epitran/PanPhon support (testing the full Teacher-Student pipeline), and (b) those without G2P resources (testing Student generalisation). For each query, the top 15 matches by cosine similarity were retrieved and manually evaluated for phonetic plausibility. Representative examples appear in Appendix A.

7 Discussion

7.1 Internal Validation: The Core Result

The primary contribution of this work is demonstrated on the internal validation benchmark, where the model substantially outperforms string similarity baselines operating on identical input representations. At the optimal threshold, the model achieves F1=0.991 compared to Jaro-Winkler's F1=0.949 and Levenshtein's F1=0.937—an improvement of 4–5 percentage points.

More significant than the aggregate metrics is the *separation* between positive and negative pairs. The learned embeddings push 95% of negative pairs below similarity 0.485, whereas Levenshtein negatives extend to 0.667 and Jaro-Winkler negatives reach 0.900. This separation has practical consequences: it enables confident automated matching at thresholds where baseline methods would produce unacceptable false positive rates. For large-scale gazetteer reconciliation involving millions of candidate pairs, this margin is essential.

7.2 The Generalist-Specialist Trade-off

The MEHDIE benchmark results reveal a fundamental tension in cross-lingual matching: models optimised for global consistency necessarily sacrifice sensitivity to language-family-specific patterns.

The MEHDIE system encodes explicit knowledge of Semitic consonantal roots, correctly treating radical vowel alternations (e.g., Arabic *kitāb* / Hebrew *ketav*) as identity-preserving. This model, trained on global data where vowel changes typically signal distinct entities (English *London* vs. *Linden*), learns a more conservative similarity function. This explains the performance gap on TS7 and TS9.

Notably, the model does not consistently outperform simple Levenshtein on MEHDIE testsets when both operate on AnyAscii romanization (average F-5: 0.616 vs. 0.651). However, this comparison is instructive: the model achieves *higher recall* on most testsets (e.g., TS9: 100% vs. 88.9%; TS11: 96.9% vs. 93.8%) at the cost of lower precision. For candidate generation workflows where human reviewers filter false positives, this trade-off may be preferable to the higher-precision, lower-recall profile of edit distance.

The gap between MEHDIE's engineered baselines (average F-5: 0.76) and the AnyAscii baselines (0.65) quantifies the value of linguistic preprocessing for this domain. The learned model cannot fully compensate for missing linguistic structure in its input representation. This suggests that for specialised domains like Semitic cross-script matching, combining learned embeddings with linguistically informed preprocessing may yield the best results.

7.3 Embedding Space Characteristics

The 64-dimensional embedding space appears sufficient for the current task, but the 14% of positive pairs scoring below 0.9 similarity suggests headroom for improvement. These difficult cases warrant investigation: preliminary analysis indicates they cluster around (a) extreme historical drift (e.g., Latin to modern vernacular), (b) competing transliteration standards, and (c) borrowed names with phonological adaptation.

Increasing embedding dimensionality to 128 or 256 dimensions might capture more nuanced phonetic distinctions, but would increase computational cost both during training and at inference time when input strings must be encoded for search. A careful trade-off analysis would be required before deployment.

7.4 Limitations

7.4.1 Grapheme-to-Phoneme Coverage

Epitran supports approximately 100 languages, but historically significant languages (Middle Persian, Classical Tibetan, pre-modern Korean) lack resources. The Student encoder provides partial mitigation through transfer from related scripts, but truly out-of-distribution writing systems (cuneiform, hieroglyphics) would require dedicated G2P development or transliteration to supported scripts.

7.4.2 Pronunciation Variation

The model assumes canonical pronunciation, but toponyms often have multiple valid realisations (regional accents, historical shifts, spelling pronunciations). Future work could explore multi-embedding approaches or uncertainty quantification.

7.4.3 Phonetic vs. Geographic Disambiguation

The model cannot distinguish phonetically identical but geographically distinct places (Paris, Texas vs. Paris, France). Integration with geographic features in the WHG PLACE pipeline addresses this limitation at the system level.

7.4.4 Input Representation

The MEHDIE results demonstrate that naive romanization (AnyAscii) discards phonetic information that linguistically informed transliteration preserves. For cross-script matching between related languages (e.g., Hebrew-Arabic), the model's input representation may be a limiting factor independent of architectural capacity.

7.5 Future Directions

Several extensions merit investigation:

- **Semitic fine-tuning**: Incorporating MEHDIE benchmark data as curriculum training examples could improve performance on morphologically complex language pairs without sacrificing global precision.
- **Improved romanization**: Replacing AnyAscii with linguistically informed transliteration for specific script pairs could recover phonetic information currently lost in preprocessing.
- **Language and script identification**: Automatic detection of source language could improve Student encoder performance, particularly for scripts where the same character sequences have different phonetic realisations across languages.
- **Semantic encoding**: Classifying toponymic elements (e.g., "-bury", "-chester", "Saint-") to capture geographic concepts could complement phonetic similarity with semantic features, though this remains at the conceptual stage.
- **Attention interpretability**: Visualising learned attention weights would likely reveal which phonetic segments carry discriminative information—one might expect consonant clusters and word-initial segments to receive higher weight, as these are typically most stable across transliteration systems.

8 Conclusion

I have presented a neural architecture for cross-lingual toponym matching that learns phonetic similarity from articulatory features. On the internal validation benchmark, the model substantially outperforms string similarity baselines (F1=0.991 vs. 0.949), with learned embeddings achieving markedly better separation between matching and non-matching pairs than edit-distance metrics.

On the external MEHDIE benchmark for Semitic cross-script matching, results are mixed: the model approaches or exceeds baseline performance on testsets involving European-Semitic pairs (TS11: 0.857) but underperforms on Hebrew-Arabic pairs where specialised linguistic knowledge provides significant advantage. This highlights a fundamental trade-off between global consistency and language-family-specific flexibility.

The model offers a scalable, language-agnostic approach to phonetic similarity that requires no expert rule-crafting and operates efficiently at scale. Embeddings for 68.7 million toponyms have been computed and deployed in the World Historical Gazetteer's reconciliation infrastructure. The trained model and code are released to facilitate further research in global digital humanities.

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A Qualitative Evaluation Tables

Series 1: Epitran-supported languages

Table 4: Nearest neighbours for ⊠⊠⊠ [zh] (Epitran-supported)

Rank	Score	Name	Lang	IPA
Q	–	⊠⊠⊠	zh	ukueit^{h}ian
1	0.9999	⊠⊠⊠	zh	p^{h}anant͡s^{h}un
2	0.9999	⊠⊠⊠	zh	kait͡ɕiaʈ͡ʂuaŋ
3	0.9999	⊠⊠⊠	zh	p^{h}ananʈ͡ʂen
4	0.9999	⊠⊠⊠	zh	ʂeŋ□ʂɻ̩
5	0.9999	⊠⊠⊠	zh	ʂeŋ□ʂɻ̩
6	0.9999	⊠⊠⊠	zh	kait͡ɕiat͡s^{h}un
7	0.9999	⊠⊠⊠	zh	ʂeŋ□ʂɻ̩
8	0.9999	⊠⊠⊠	zh	p^{h}ananʈ͡ʂen
9	0.9999	⊠⊠⊠	zh	p^{h}anfut͡s^{h}un
10	0.9999	⊠⊠⊠	zh	ʂeŋɕiaŋt͡s^{h}un
11	0.9999	⊠⊠⊠	zh	ʂeŋp^{h}iŋʈ͡ʂan
12	0.9999	⊠⊠⊠	zh	p^{h}anʂanɕian
13	0.9999	⊠⊠⊠	zh	p^{h}anʂant^{h}ou
14	0.9999	⊠⊠⊠	zh	ʂeŋʈ͡ʂuaŋt͡s^{h}un
15	0.9999	⊠⊠⊠	zh	ʂeŋʈ͡ʂuaŋʈ͡ʂan

Series 2: Non-Epitran languages

Queries in Cyrillic script

Queries in Greek script

Queries in Arabic script

Queries in Hebrew script

Queries in Cjk script

Table 77: Cross-script evaluation summary

Metric	Value
Query toponyms evaluated	50
Neighbours per query	15
Scripts encountered	9
Avg. cross-script neighbours	10.7 / 15
Cross-script neighbour rate	71.3%
Mean score (cross-script)	0.9999
Mean score (same-script)	0.9988

Table 5: Nearest neighbours for **Othón P. Blanco** [pt] (Epitran-supported)

Rank	Score	Name	Lang	IPA
Q	—	Othón P. Blanco	pt	otɔn p. blɐnko
1	0.9969	Othón P. Blanco	it	oton p. blanko
2	0.9955	Othón P. Blanco	es	oton p. blanko
3	0.9935	Nhlangenyuke	und	—
4	0.9931	Domaine-Bellevue	und	—
5	0.9929	Dabun Gelang	nan	—
6	0.9929	Dabun Gelang	min	—
7	0.9929	Dabun Gelang	su	—
8	0.9928	Dabun Gelang	map	—
9	0.9918	Na Mlynček	und	—
10	0.9916	Monte Malaonga	und	—
11	0.9914	RPTRA Malinjo	und	—
12	0.9913	Gunung Walana	id	ɡunuŋ walana
13	0.9913	Khlong Lam Nam	und	—
14	0.9913	Khlong Lam Nam	ceb	—
15	0.9911	Khlong Nam Dam	ceb	—

Table 6: Nearest neighbours for **Darreh-ye Karāshk** [sv] (Epitran-supported)

Rank	Score	Name	Lang	IPA
Q	—	Darreh-ye Karāshk	sv	dɐreːh-yːeː kɑːrɑ̄shk
1	0.9933	Darreh-ye Zereshk	sv	dɐreːh-yːeː seːrɛshk
2	0.9911	Darreh-ye Garashkeh	sv	dɐreːh-yːeː ɡɑːrɐshɕeːh
3	0.9909	North Terrace Creek	und	—
4	0.9906	North Terrace Creek	ceb	—
5	0.9903	Cerro Jaquiracirca	en	sɛɹow d͡ʒækwɹæsɹ̩kə
6	0.9903	Darreh-ye Veresk	sv	dɐreːh-yːeː veːrɛɧ
7	0.9898	Darreh-ye Kharkesh	ceb	—
8	0.9895	Great Marsh Creek	ceb	—
9	0.9893	Darreh-ye Kharkesh	und	—
10	0.9892	Great Marsh Creek	und	—
11	0.9883	Darreh-ye Lā-ye Ḩorkesh	und	—
12	0.9880	Harish Mukherjee Park	en	hɛɹɪʃ məkhɹ̩d͡ʒi pɑɹk
13	0.9876	Darreh-ye ʻArūsak	sv	dɐreːh-yːeː ʻɑːrʉ̄sɑːk
14	0.9875	Cherokee Trace	und	—
15	0.9875	North Terrace Park	ceb	—

Table 7: Nearest neighbours for **The Digger Farm** [en] (Epitran-supported)

Rank	**Score**	**Name**	**Lang**	**IPA**
Q	—	The Digger Farm	en	ðə dɪgɹ̩ faɹm
1	0.9928	High Park Farm	en	haj pɑɹk fɑɹm
2	0.9927	De Jager Farm	und	—
3	0.9891	Liegaxia'erma	und	—
4	0.9876	New York Farm	en	nu jɔɹk fɑɹm
5	0.9873	Bakeapple Mash	en	bejkpəl mæʃ
6	0.9867	Deer Lake Farm	en	dɪɹ lejk fɑɹm
7	0.9866	The Park Farm	en	ðə pɑɹk fɑɹm
8	0.9866	Takat Mariam	en	tɑkət mɛɹiæm
9	0.9864	New Park Farm	en	nu pɑɹk fɑɹm
10	0.9861	Dickerage Farm	en	dɪkɹ̩ɪd͡ʒ fɑɹm
11	0.9858	GBPS Turk Farm	en	gbps tɹ̩k fɑɹm
12	0.9853	Lower Vagg Farm	en	lowɹ̩ vægg fɑɹm
13	0.9839	Nagareha Yama	und	—
14	0.9838	Kūh-e Rāq Charmī	nl	kȳh-ɛ rāk xɑrmī
15	0.9834	Wick Bridge Farm	en	wɪk bɹɪd͡ʒ fɑɹm

Table 8: Nearest neighbours for **Svirok** [ro] (Epitran-supported)

Rank	**Score**	**Name**	**Lang**	**IPA**
Q	—	Svirok	ro	svirok
1	0.9969	Svirok	uk	svirok
2	0.9951	Ivyrock	en	ajvɪɹɑk
3	0.9913	Goworek	ceb	—
4	0.9909	Svirok	en	svɪɹɑk
5	0.9904	Kwirok	id	kwirok
6	0.9898	Swirok	de	zviːroːk
7	0.9896	Noveruk	und	—
8	0.9894	Keverok	fr	kəvəʀɔk
9	0.9886	Goworek	lld	—
10	0.9886	Goworek	und	—
11	0.9884	Siverek	en	sɪvɹ̩ɪk
12	0.9884	Overock	en	owvɹɑk
13	0.9884	Voorwyk	und	—
14	0.9884	Keverok	de	kɛːfɛːroːk
15	0.9865	Gourock	pt	gowuɾokk

Table 9: Nearest neighbours for **Кынок** [kk] (Epitran-supported)

Rank	Score	Name	Lang	IPA
Q	–	Кынок	kk	kənok
1	0.9999	Грэйс	kk	grejs
2	0.9999	Джепт	kk	dʒjept
3	0.9999	Джепт	kk	dʒjept
4	0.9999	Гуапо	kk	gwɑpo
5	0.9999	Джово	kk	dʒovo
6	0.9999	Джойс	kk	dʒojs
7	0.9999	Джойс	kk	dʒojs
8	0.9999	Джоли	kk	dʒolij
9	0.9999	Джонс	kk	dʒons
10	0.9999	Драге	kk	drɑgje
11	0.9999	Дибах	kk	dijbɑχ
12	0.9999	Гурен	kk	gwrjen
13	0.9999	Дивля	kk	dijvljɑ
14	0.9999	Гусев	kk	gwsjev
15	0.9999	Димон	kk	dijmon

Table 10: Nearest neighbours for ⊠⊠⊠⊠⊠ [fa] (Epitran-supported)

Rank	Score	Name	Lang	IPA
Q	–	⊠⊠⊠⊠⊠	fa	ɒfhlr
1	1.0000	⊠⊠⊠⊠⊠⊠	fa	vɒkɒsɒ
2	1.0000	⊠⊠⊠⊠⊠⊠	fa	vɒkɒsv
3	1.0000	⊠⊠⊠⊠⊠⊠	fa	vɒrtsɒ
4	1.0000	⊠⊠⊠⊠⊠⊠	fa	vɒjljɒ
5	1.0000	⊠⊠⊠⊠⊠⊠	fa	vɒjmɒr
6	1.0000	⊠⊠⊠⊠⊠⊠	fa	vrɣvjh
7	1.0000	⊠⊠⊠⊠⊠⊠	fa	vɒrslj
8	1.0000	⊠⊠⊠⊠⊠⊠	fa	vrnvsj
9	1.0000	⊠⊠⊠⊠⊠⊠	fa	vrnvks
10	1.0000	⊠⊠⊠⊠⊠⊠	fa	vrnjɒb
11	1.0000	⊠⊠⊠⊠⊠⊠	fa	vtrnjk
12	1.0000	⊠⊠⊠⊠⊠⊠	fa	vtjmnt
13	1.0000	⊠⊠⊠⊠⊠⊠	fa	vrvndj
14	1.0000	⊠⊠⊠⊠⊠	fa	vsjsr
15	1.0000	⊠⊠⊠⊠⊠⊠	fa	vʃtrɒv

Table 11: Nearest neighbours for **Katelijnevest** [nl] (Epitran-supported)

Rank	Score	Name	Lang	IPA
Q	—	Katelijnevest	nl	kaːteːlɛjneːvɛst
1	0.9914	Adelinental	und	—
2	0.9907	Le Oendulamuli	id	lə oəndulamuli
3	0.9904	Beedle Windmill	und	—
4	0.9901	Field Windmill	und	—
5	0.9899	Adelinenthal	und	—
6	0.9897	Taltanzoukht	und	—
7	0.9897	Chandler Island	sv	ɧɐndleːr ɪslɐnd
8	0.9881	Katelijnevest	en	kætlid͡ʒnɛvɛst
9	0.9881	Shetlandeiland	nl	ʃɛtlɑndɛjɪlɑnt
10	0.9880	Basti Bandwali	und	—
11	0.9879	Stalinstadt	ga	—
12	0.9877	Basti Jandwali	und	—
13	0.9874	Elton,nedre	no	—
14	0.9872	Celtnieku iela	nl	sɛltniːɛkʏ iːɛlaː
15	0.9872	Odlingshult	und	—

Table 12: Nearest neighbours for **Ouâdi ed Dîk** [sv] (Epitran-supported)

Rank	Score	Name	Lang	IPA
Q	—	Ouâdi ed Dîk	sv	uːʉːɑ̂ːdiː eːd dîːk
1	0.9938	Sidi Moatek	und	—
2	0.9929	Mota Wedik	id	mota wədik
3	0.9916	Goode Heights	und	—
4	0.9915	Voydatishki	und	—
5	0.9908	Noaidegeađgi	nn	—
6	0.9907	Koudiat Biekh	en	kawdiæt bik
7	0.9906	Ševet Cedek	cs	ʃevet t͡sedek
8	0.9905	Noaidegeađgi	se	—
9	0.9905	Noaidegeađgi	nb	—
10	0.9902	Ouadi Michtâq	sv	uːʉːɑːdiː mɪɧtɑ̂ːq
11	0.9901	Oued Boitiek	ceb	—
12	0.9901	Oued Boitiek	und	—
13	0.9898	Oued Ouddigh	und	—
14	0.9897	Nooitgedeg	und	—
15	0.9897	Oued Ouddigh	ceb	—

Table 13: Nearest neighbours for **Iggui Ouferni** [fi] (Epitran-supported)

Rank	**Score**	**Name**	**Lang**	**IPA**
Q	—	Iggui Ouferni	fi	iggui ouferni
1	0.9958	Agouni Ifri	sv	ɑːguːʉːniː ɪfriː
2	0.9929	Ropica Górna	et	ropːicɑ kórnɑ
3	0.9911	Çiçekpınarı	pt	sisɛkpɪnɐɾɪ
4	0.9911	Çiçekpınarı	pt	sisɛkpɪnɐɾɪ
5	0.9875	Agouni Ifri	ast	—
6	0.9875	Agouni Ifri	und	—
7	0.9866	Setjopereng	und	—
8	0.9859	Lgota Gorna	sq	lgɔta gɔɾna
9	0.9852	Rypetjern	nb	—
10	0.9852	Rypetjern	no	—
11	0.9851	Rushy Spring	ceb	—
12	0.9851	Rushy Spring	und	—
13	0.9850	Raptjørna	no	—
14	0.9850	Raptjørna	und	—
15	0.9849	Ridge Spring	tr	ɾidge spɾing

Table 14: Summary statistics for Epitran-supported languages ($n = 10$, $k = 15$)

Metric	**Value**
Mean similarity score	0.9926
Min similarity score	0.9834
Max similarity score	1.0000
Avg. same-language neighbours	6.3 / 15
Avg. neighbours with IPA	9.1 / 15

Table 15: Nearest neighbours for **Saint-Joseph-de-Rivière** [scn] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	Saint-Joseph-de-Rivière	scn	—
1	0.9999	Saint-Joseph-de-Rivière	br	—
2	0.9999	Saint-Joseph-de-Rivière	eo	—
3	0.9999	Saint-Joseph-de-Rivière	fo	—
4	0.9999	Saint-Joseph-de-Rivière	frc	—
5	0.9999	Saint-Joseph-de-Rivière	gsw	—
6	0.9999	Saint-Joseph-de-Rivière	ie	—
7	0.9999	Saint-Joseph-de-Rivière	kg	—
8	0.9999	Saint-Joseph-de-Rivière	kl	—
9	0.9999	Saint-Joseph-de-Rivière	la	—
10	0.9999	Saint-Joseph-de-Rivière	lmo	—
11	0.9999	Saint-Joseph-de-Rivière	mul	—
12	0.9999	Saint-Joseph-de-Rivière	nap	—
13	0.9999	Saint-Joseph-de-Rivière	pap	—
14	0.9999	Saint-Joseph-de-Rivière	pcd	—
15	0.9999	Saint-Joseph-de-Rivière	rgn	—

Table 16: Nearest neighbours for **Córrego Riachão** [und] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	Córrego Riachão	und	—
1	0.9997	Córrego Riachão	ceb	—
2	0.9932	Córrego Caicaí	pt	kɔʁɛgo kajkaj́
3	0.9925	Barrio Tayacaja	und	—
4	0.9923	Córrego Baiaca	und	—
5	0.9922	Córrego Baiaca	pt	kɔʁɛgo bɐjɐkɐ
6	0.9916	Ery-Makouguié 2	fr	əʀi-makugje 2
7	0.9913	Córrego Bahia	und	—
8	0.9905	Córrego Chico	und	—
9	0.9905	Barrio Macuathi	sh	—
10	0.9905	Barrio Macuathi	und	—
11	0.9904	Ery-Makouguié 1	fr	əʀi-makugje 1
12	0.9900	Cerro Ereo Chico	nl	sɛrrɔ eːrɛɔ xɪcoː
13	0.9898	Arroyo Caña Seca	nl	ɑrrɔjɔ cɑnjɑ sɛcaː
14	0.9896	Córrego Riacho	und	—
15	0.9894	Córrego Taioca	pt	kɔʁɛgo tɐjokɐ

Table 17: Nearest neighbours for **Beni Boukitoun** [ceb] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	Beni Boukitoun	ceb	—
1	0.9995	Beni Boukitoun	und	—
2	0.9932	Tanjung Lautang	id	tand͡ʑuŋ lautaŋ
3	0.9926	Bukit Lentang	ceb	—
4	0.9926	Bukit Sentang	und	—
5	0.9926	Bukit Lentang	und	—
6	0.9905	Saunkt Poitn	bar	—
7	0.9904	Tuco et Cantaou	und	—
8	0.9899	Tunjungan Wetan	id	tund͡ʑuŋan wətan
9	0.9889	Puchong Intan	und	—
10	0.9889	Cnoc Sciatháin	ga	—
11	0.9888	Rio do Banco	und	—
12	0.9885	Punta Fixón	und	—
13	0.9885	Punta Pixín	und	—
14	0.9884	Gonnoscodina	sq	gɔnnɔst͡sɔdina
15	0.9883	Sawangan Wetan	es	sawangan wetan

Table 18: Nearest neighbours for **Ozero Isayevo** [und] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	Ozero Isayevo	und	—
1	0.9911	Casa Rosario 4	es	kasa ɾosaɾjo 4
2	0.9889	Casa Rosario 9	es	kasa ɾosaɾjo 9
3	0.9888	Ozero Buzayevo	ceb	—
4	0.9888	Ozero Buzayevo	und	—
5	0.9881	Casa Rosario 6	es	kasa ɾosaɾjo 6
6	0.9877	Ozero Dausu-Nov	und	—
7	0.9873	Ozero Shishevo	ceb	—
8	0.9869	Ozero Shishevo	und	—
9	0.9853	Caserio Zuatzu	es	kaseɾjo swat͡su
10	0.9851	Ozero Ty-Neyso-To	und	—
11	0.9849	Ozero Ngosavey	und	—
12	0.9847	Ozero Levashevo	ceb	—
13	0.9847	Kisgruveåsen	en	kɪsgɹuvåsɛn
14	0.9846	Casa Rosario 15	es	kasa ɾosaɾjo 15
15	0.9845	Ozero Peschane	und	—

Table 19: Nearest neighbours for **Шевкатие** [tt] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	Шевкатие	tt	—
1	1.0005	GÜLERMAK	und	—
2	1.0005	KAIPADAR	und	—
3	1.0005	KALAWEIH	und	—
4	1.0005	KANDHIYA	und	—
5	1.0005	KARABIRI	und	—
6	1.0005	KARADIGA	und	—
7	1.0005	KATHIARA	und	—
8	1.0005	KAYADERE	und	—
9	1.0005	KHAROUDI	und	—
10	1.0005	KJELLAND	nb	—
11	1.0005	KM⊠⊠⊠⊠⊠⊠	und	—
12	1.0005	KOUBANKO	und	—
13	1.0005	KOROGORO	und	—
14	1.0005	KOSSOBIO	und	—
15	1.0005	KOSTVEIT	nb	—

Table 20: Nearest neighbours for **Emeşe** [nb] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	Emeşe	nb	—
1	0.9999	Emeşe	ceb	—
2	0.9999	Emeşe	da	—
3	0.9999	Emeşe	eu	—
4	0.9999	Emeşe	mul	—
5	0.9999	Emeşe	sk	—
6	0.9999	Emeşe	war	—
7	0.9998	Emeşe	nn	—
8	0.9998	Emeşe	und	—
9	0.9998	Emeşe	oc	—
10	0.9998	Emeşe	sl	—
11	0.9925	Tamahú	sv	tɑːmɑːhʉ́ː
12	0.9900	Ramaḍah	und	—
13	0.9898	Jamaşah	und	—
14	0.9897	Hameh Ja	nan	—
15	0.9891	Ammaja	fi	ɑm:ɑjɑ

Table 21: Nearest neighbours for **1908ko Udako Olinpiar Jokoak** [eu] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	1908ko Udako Olinpiar Jokoak	eu	—
1	0.9913	1992ko Udako Olinpiar Jokoak	eu	—
2	0.9888	1940ko Udako Olinpiar Jokoak	eu	—
3	0.9886	1980ko Udako Olinpiar Jokoak	eu	—
4	0.9870	1928ko Neguko Olinpiar Jokoak	eu	—
5	0.9840	1992ko Neguko Olinpiar Jokoak	eu	—
6	0.9832	1960ko Neguko Olinpiar Jokoak	eu	—
7	0.9823	1994ko Neguko Olinpiar Jokoak	eu	—
8	0.9818	1936ko Udako Olinpiar Jokoak	eu	—
9	0.9813	parc national du Lac Nakuru	fr	paʀk nasjɔnal dy lak nakyʀy
10	0.9792	most přes Rokytku v ulici Čelákovická	nl	mɔst přɛs rɔkjtkʏ v yːliːsɪ čeːlɑ́koːvɪkɑ́
11	0.9757	2014ko Neguko Olinpiar Jokoak	eu	—
12	0.9747	Centro de Eventos Valle del Pacífico	es	sentɾo de ebentos baʝe del pasifiko
13	0.9746	3 Botaniczna Street in Kraków	en	3 bətænɪt͡ʃnə stɹit ɪn kɹækódʌbəlju
14	0.9745	převod ze Sloupnického potoka	en	piřɛvɑd zi slupnɪkéhow pətowkə
15	0.9744	2010eko Neguko Olinpiar Jokoak	eu	—

Table 22: Nearest neighbours for **Kauvbergje** [ceb] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	Kauvbergje	ceb	—
1	0.9976	Kauvbergje	no	—
2	0.9976	Kauvbergje	und	—
3	0.9922	Afo Market	yo	afo market
4	0.9911	Sophia Hirsch	und	—
5	0.9909	Appo Keriche	und	—
6	0.9901	Do vrbíc	und	—
7	0.9893	Taopa Barat	en	towpə bɑɹət
8	0.9893	Taopa Barat	en	towpə bɑɹət
9	0.9885	Popa Tērce	und	—
10	0.9869	Bayou Lafourche	en	baju lɑfɔɹt͡ʃ
11	0.9857	Pawpaw Beach	und	—
12	0.9857	Hope Acres	und	—
13	0.9852	Bay Roberts	pt	bɐj ɾobɛrtʃ
14	0.9850	Nuba-Berge	de	nuːbaː-bɛːrgə
15	0.9849	Kuupahaa Gulch	ceb	—

Table 23: Nearest neighbours for **Marškalni** [und] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	Marškalni	und	—
1	0.9944	Karklinai	nan	—
2	0.9943	Karklyni	und	—
3	0.9940	Karklinai	und	—
4	0.9939	Sarkhanlŭ	und	—
5	0.9939	Markhanlă	und	—
6	0.9936	Karklenay	und	—
7	0.9935	Krogkalni	und	—
8	0.9934	Garīkolni	und	—
9	0.9925	Margalanj	und	—
10	0.9925	Margalanj	hy	—
11	0.9923	Qaraghānlu	und	—
12	0.9914	Karkliny	und	—
13	0.9913	Garkolni	und	—
14	0.9910	Baragan Well	und	—
15	0.9909	Dragon Hall	und	—

Table 24: Nearest neighbours for ⊠⊠⊠⊠⊠⊠2⊠ [und] (no Epitran)

Rank	Score	Name	Lang	IPA
Q	—	⊠⊠⊠⊠⊠⊠2⊠	und	—
1	0.9998	⊠⊠⊠⊠⊠⊠2⊠	und	—
2	0.9998	⊠⊠⊠⊠⊠⊠2⊠	und	—
3	0.9998	⊠⊠·⊠⊠⊠2⊠	und	—
4	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
5	0.9997	⊠⊠⊠⊠⊠B2⊠	und	—
6	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
7	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
8	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
9	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
10	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
11	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
12	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
13	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
14	0.9997	⊠⊠⊠⊠⊠⊠2⊠	und	—
15	0.9997	⊠⊠·⊠⊠⊠2⊠	und	—

Table 25: Summary statistics for Non-Epitran languages ($n = 10$, $k = 15$)

Metric	Value
Mean similarity score	0.9930
Min similarity score	0.9744
Max similarity score	1.0005
Avg. same-language neighbours	5.2 / 15
Avg. neighbours with IPA	3.0 / 15

Table 26: Comparison of embedding quality: Epitran-supported vs non-Epitran languages

Metric	Epitran	Non-Epitran
Mean similarity (all neighbours)	0.9926	0.9930
Mean similarity (top-1)	0.9961	0.9974
Mean similarity (top-5)	0.9942	0.9953

Table 27: Cross-script neighbours for **Ирэгандя** [und, cyrillic] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Ирэгандя	und	cyrillic	—
1	1.0005	GÜLERMAK	und	latin	—
2	1.0005	KAIPADAR	und	latin	—
3	1.0005	KALAWEIH	und	latin	—
4	1.0005	KANDHIYA	und	latin	—
5	1.0005	KARABIRI	und	latin	—
6	1.0005	KARADIGA	und	latin	—
7	1.0005	KATHIARA	und	latin	—
8	1.0005	KAYADERE	und	latin	—
9	1.0005	KHAROUDI	und	latin	—
10	1.0005	KJELLAND	nb	latin	—
11	1.0005	KM⊠⊠⊠⊠⊠⊠	und	latin	—
12	1.0005	KOUBANKO	und	latin	—
13	1.0005	KOROGORO	und	latin	—
14	1.0005	KOSSOBIO	und	latin	—
15	1.0005	KOSTVEIT	nb	latin	—

Table 28: Cross-script neighbours for **Кревишон** [ru, cyrillic] (0/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Кревишон	ru	cyrillic	krʲevʲiʂon
1	0.9998	Джизасай	ru	cyrillic	d͡ʒizasaj
2	0.9998	Джиламыш	ru	cyrillic	d͡ʒilamɨʂ
3	0.9998	Джиланды	ru	cyrillic	d͡ʒilandɨ
4	0.9998	Джилимпа	ru	cyrillic	d͡ʒilʲimpa
5	0.9998	Джилинка	ru	cyrillic	d͡ʒilʲinka
6	0.9998	Джилихур	ru	cyrillic	d͡ʒilʲixur
7	0.9998	Джялокан	ru	cyrillic	d͡ʒalokan
8	0.9998	Джёнкюке	ru	cyrillic	d͡ʒonkʲukʲe
9	0.9998	Джильгад	ru	cyrillic	d͡ʒilʲgad
10	0.9998	Дзамараш	ru	cyrillic	dzamaraʂ
11	0.9998	Дзангдог	ru	cyrillic	dzangdog
12	0.9998	Дзаплаки	ru	cyrillic	dzaplakʲi
13	0.9998	Джоджора	ru	cyrillic	d͡ʒod͡ʒora
14	0.9998	Джолондо	ru	cyrillic	d͡ʒolondo
15	0.9998	Джолукax	ru	cyrillic	d͡ʒolukax

Table 29: Cross-script neighbours for **Карабаны** [und, cyrillic] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Карабаны	und	cyrillic	—
1	1.0005	GÜLERMAK	und	latin	—
2	1.0005	KAIPADAR	und	latin	—
3	1.0005	KALAWEIH	und	latin	—
4	1.0005	KANDHIYA	und	latin	—
5	1.0005	KARABIRI	und	latin	—
6	1.0005	KARADIGA	und	latin	—
7	1.0005	KATHIARA	und	latin	—
8	1.0005	KAYADERE	und	latin	—
9	1.0005	KHAROUDI	und	latin	—
10	1.0005	KJELLAND	nb	latin	—
11	1.0005	KM⊠⊠⊠⊠⊠⊠	und	latin	—
12	1.0005	KOUBANKO	und	latin	—
13	1.0005	KOROGORO	und	latin	—
14	1.0005	KOSSOBIO	und	latin	—
15	1.0005	KOSTVEIT	nb	latin	—

Table 30: Cross-script neighbours for **Казанский аэропорт** [ru, cyrillic] (0/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Казанский аэропорт	ru	cyrillic	kazanskʲij aeroport
1	0.9997	Догзмакин Калмыкия	ru	cyrillic	dogzmakʲin kalmɨkʲia
2	0.9997	Заставный переулок	ru	cyrillic	zastavnɨj pʲerʲeulok
3	0.9997	Здоровому человеку	ru	cyrillic	zdorovomu t͡ɕʲelovʲeku
4	0.9997	Заводская больница	ru	cyrillic	zavodskaa bolʲnit͡sa
5	0.9997	Заводский переулок	ru	cyrillic	zavodskʲij pʲerʲeulok
6	0.9997	Загорское кладбище	ru	cyrillic	zagorskoe kladbʲiɕːe
7	0.9997	Заозёрное кладбище	ru	cyrillic	zaozʲornoe kladbʲiɕːe
8	0.9997	Дубенский поссовет	ru	cyrillic	dubʲenskʲij possovʲet
9	0.9997	Евлахский аэропорт	ru	cyrillic	jevlaxskʲij aeroport
10	0.9997	Еврейская синагога	ru	cyrillic	jevrʲejskaa sʲinagoga
11	0.9997	Дунайский проспект	ru	cyrillic	dunajskʲij prospʲekt
12	0.9997	Дыренская Кактолга	ru	cyrillic	dɨrʲenskaa kaktolga
13	0.9997	Заводская Камбарка	ru	cyrillic	zavodskaa kambarka
14	0.9997	Неманский переулок	ru	cyrillic	nemanskʲij pʲerʲeulok
15	0.9997	Новинский переулок	ru	cyrillic	novʲinskʲij pʲerʲeulok

Table 31: Cross-script neighbours for **Кътинска река** [bg, cyrillic] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Кътинска река	bg	cyrillic	—
1	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	und	cjk	—
2	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	und	latin	—
3	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	und	cjk	—
4	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	und	cjk	—
5	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	und	cjk	—
6	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	ber	latin	—
7	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	syl	latin	—
8	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	syl	latin	—
9	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	mni	latin	—
10	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	und	cjk	—
11	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	und	georgian	—
12	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	xmf	georgian	—
13	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	sat	latin	—
14	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	mnw	latin	—
15	1.0000	⌧⌧⌧⌧⌧⌧⌧⌧ ⌧⌧⌧⌧	und	cjk	—

Table 32: Cross-script neighbours for **Ишкошим** [cs, cyrillic] (13/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Ишкошим	cs	cyrillic	—
1	0.9995	Γρεβενά	cs	greek	γρ□β□□ά
2	0.9995	HELLADA	cs	latin	ɦellada
3	0.9995	BAJKING	cs	latin	bajkɪnk
4	0.9995	CALTECH	cs	latin	t͡saltex
5	0.9995	⌧⌧⌧⌧⌧⌧⌧	cs	latin	—
6	0.9995	Ямантау	cs	cyrillic	—
7	0.9995	Μέτσοβο	cs	greek	μ□□σ□β□
8	0.9995	MPSVRSR	cs	latin	mpsvrzr
9	0.9994	Цернина	cs	cyrillic	—
10	0.9994	⌧⌧⌧⌧⌧⌧	cs	arabic	—
11	0.9994	HOMMAGE	cs	latin	ɦommage
12	0.9994	EUROBIT	cs	latin	eurobɪt
13	0.9994	DONTИOD	cs	latin	dontиot
14	0.9994	MICHAEL	cs	latin	mɪxael
15	0.9994	FRANTIQ	cs	latin	vrancɪq

Table 33: Cross-script neighbours for **Падмуркі касцёла Звеставання Найсвяцейшай Панны Марыі 1751 г. і манастыра брыгітак 1623 г.** [und, cyrillic] (4/15 cross-script)

Rank	Score	Name
Q	—	Падмуркі касцёла Звеставання Найсвяцейшай Панны Марыі 1751 г. і манастыра брыгітак 162
1	1.0000	Падмуркі касцёла Звеставання Найсвяцейшай Панны Марыі 1751 г. і манастыра брыгітак 162
2	0.9994	[illegible]
3	0.9992	Інстытут геалогіі Карэльскага навуковага цэнтра РАН
4	0.9986	Робоча територія розсаднику декоративних рослин ГАРДИ
5	0.9985	Словакия фәннәр академиясе каршында оештырылган Людовит Штур исемендәге тел белү
6	0.9983	[illegible]
7	0.9980	[illegible]
8	0.9980	Унутраныя войскі Міністэрства ўнутраных спраўаў Рэспублікі Беларусь
9	0.9979	Памятник павшим воинам — работникам химического завода в годы ВОВ
10	0.9979	Унутраныя войскі Міністэрства ўнутраных спраўаў Беларусі
11	0.9976	Памятник месту рождения государственнго деятеля Российской Империи графа Алексея Ан
12	0.9974	Дизайн æмæ технологиты мæскуыйаг паддзахадон университет
13	0.9974	Һөйөүҙе юғалтып мөнәсәбәттәрҙе өҙөүҙе күрһәткән музей
14	0.9974	Ανώτατη Σχολή Τηλεόρασης και Κινηματογράφου του Μονάχου
15	0.9973	Усадьба семьи декабриста Александра Михайловича Каховского

Table 34: Cross-script neighbours for **Berezovyi Hrud** [fi, latin] (0/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Berezovyi Hrud	fi	latin	beretsoʋyi hrud
1	0.9935	Listrac-de-Durèze	fi	latin	listrɑk-de-durètse
2	0.9934	Sebyar Rejosari	id	latin	səbjar rəd͡ʐosari
3	0.9923	Distretto di Corosha	it	latin	distretːo di korosa
4	0.9918	Théâtre de Bressuire	fr	latin	teatʀə də bʀɛsɥiʀ
5	0.9910	Izvorski Potok	sv	latin	ɪsvɔrsɕiː puːtuːk
6	0.9909	Bērziņu Karjers	lv	latin	bɛːrziɲu karjærs
7	0.9909	Ostrykół Dworski	cs	latin	ozdrɪkoːł dworskɪ
8	0.9908	Vrieswijkstraat	de	latin	frisvɪjkstraːt
9	0.9908	Berezovii Grud	tr	latin	beɾezovii gɾud
10	0.9907	Frieswijkstraat	de	latin	frisvɪjkstraːt
11	0.9905	Berezovyj Hrud	cs	latin	berezovɪj ɦrut
12	0.9903	Listrac-de-Durèze	ro	latin	listrak-de-durèze
13	0.9898	Listrac-de-Durèze	de	latin	lɪstraːk-dɛː-duːrɛːt͡sə
14	0.9897	Listrac-de-Durèze	de	latin	lɪstraːk-dɛː-duːrɛːt͡sə
15	0.9897	Listrac-de-Durèze	de	latin	lɪstraːk-dɛː-duːrɛːt͡sə

Table 35: Cross-script neighbours for **Вале-Аўрына** [be, cyrillic] (14/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Вале-Аўрына	be	cyrillic	—
1	0.9999	BENI-MANZOR	und	latin	—
2	0.9999	Άλακ-ντονγκ	el	greek	—
3	0.9999	Άλιμ-ντονγκ	el	greek	—
4	0.9999	Όκοκ-ντονγκ	el	greek	—
5	0.9999	Όσαν-ντονγκ	el	greek	—
6	0.9999	Όσου-ντονγκ	el	greek	—
7	0.9999	⊠⊠⊠⊠-⊠⊠⊠⊠⊠⊠	hy	armenian	—
8	0.9999	⊠⊠⊠⊠-⊠⊠⊠⊠⊠⊠	hy	armenian	—
9	0.9999	⊠⊠⊠⊠-⊠⊠⊠⊠⊠⊠	hy	armenian	—
10	0.9999	Ҩада-Сакьан	und	cyrillic	—
11	0.9999	⊠⊠⊠⊠-⊠⊠⊠⊠⊠⊠	hy	armenian	—
12	0.9999	⊠⊠⊠⊠-⊠⊠⊠⊠⊠⊠	hy	armenian	—
13	0.9999	⊠⊠⊠⊠-⊠⊠⊠⊠⊠⊠	hy	armenian	—
14	0.9999	⊠⊠⊠⊠-⊠⊠⊠⊠⊠⊠	hy	armenian	—
15	0.9999	⊠⊠⊠⊠-⊠⊠⊠⊠⊠⊠	hy	armenian	—

Table 36: Cross-script neighbours for **Рағзов** [und, cyrillic] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Рағзов	und	cyrillic	—
1	1.0001	KABADI	und	latin	—
2	1.0001	KABETE	und	latin	—
3	1.0001	KABOBI	und	latin	—
4	1.0001	KAHRIF	und	latin	—
5	1.0001	KAKOMA	und	latin	—
6	1.0001	KAMTHE	und	latin	—
7	1.0001	KAROGO	und	latin	—
8	1.0001	KARSEC	gur	latin	—
9	1.0001	KASERE	und	latin	—
10	1.0001	KATECO	gpe	latin	—
11	1.0001	KBS⊠⊠⊠	und	latin	—
12	1.0001	KCC⊠⊠⊠	und	latin	—
13	1.0001	KCHUGR	sl	latin	—
14	1.0001	KEGGER	und	latin	—
15	1.0001	KERNEL	und	latin	—

Table 37: Cross-script neighbours for **Άνθεια Έβρου** [el, greek] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Άνθεια Έβρου	el	greek	—
1	1.0003	LITTLE RANGE	und	latin	—
2	1.0003	MOEHAU RANGE	und	latin	—
3	1.0003	MUONJO RIVER	und	latin	—
4	1.0003	MANUKA RANGE	ceb	latin	—
5	1.0003	MANUKA RANGE	und	latin	—
6	1.0002	KAKETU RANGE	und	latin	—
7	1.0002	KARANG AGUNG	und	latin	—
8	1.0002	KD⊠⊠⊠⊠ ⊠⊠⊠⊠⊠	und	latin	—
9	1.0002	KD⊠⊠⊠⊠ ⊠⊠⊠⊠⊠	und	latin	—
10	1.0002	KD⊠⊠⊠⊠ ⊠⊠⊠⊠⊠	und	latin	—
11	1.0002	KILELE MPETE	und	latin	—
12	1.0002	KNOBBY RANGE	ceb	latin	—
13	1.0002	KNOBBY RANGE	und	latin	—
14	1.0002	KTX⊠⊠⊠ ⊠⊠⊠⊠⊠	und	latin	—
15	1.0000	⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠	am	latin	jəfərənɨsaj polineʒija

Table 38: Cross-script neighbours for **Μετόχι Σταυρονικήτα** [el, greek] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Μετόχι Σταυρονικήτα	el	greek	—
1	0.9999	⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠	und	cjk	—
2	0.9999	Якутия Республикась	myv	cyrillic	—
3	0.9999	⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠	und	arabic	—
4	0.9999	Язовир Стамболийски	bg	cyrillic	—
5	0.9999	⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠	azb	arabic	—
6	0.9999	TANANA MADIROMANERA	und	latin	—
7	0.9999	SÅHEIM KRAFTSTASJON	nb	latin	—
8	0.9999	JA⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠	und	cjk	—
9	0.9999	⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠	xmf	georgian	—
10	0.9999	⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠	und	georgian	—
11	0.9999	⊠⊠⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠	xmf	georgian	—
12	0.9999	VILLAS AEROTECNICOS	und	latin	—
13	0.9999	Обшори Бамбараканда	tg	cyrillic	—
14	0.9999	Обшори Веттисфоссен	tg	cyrillic	—
15	0.9999	Обшори Виллиамспорт	tg	cyrillic	—

Table 39: Cross-script neighbours for **Διάμεσο** [el, greek] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Διάμεσο	el	greek	—
1	1.0018	ECUASAL	und	latin	—
2	1.0018	EDESCON	und	latin	—
3	1.0018	ELBISCO	und	latin	—
4	1.0018	ELHNYNA	und	latin	—
5	1.0018	ELIPSON	und	latin	—
6	1.0018	ELISAVA	eu	latin	—
7	1.0018	ELVENES	nb	latin	—
8	1.0018	ELYSIUM	und	latin	—
9	1.0018	EMASTRU	und	latin	—
10	1.0018	EMBALSE	und	latin	—
11	1.0018	EMBRAPA	und	latin	—
12	1.0018	ENCOMIL	und	latin	—
13	1.0018	ENGEVIX	und	latin	—
14	1.0018	EPITECH	pcd	latin	—
15	1.0018	ERABPRP	und	latin	—

Table 40: Cross-script neighbours for **Οικόπεδο Χαραγκιωνη** [und, greek] (14/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Οικόπεδο Χαραγκιωνη	und	greek	—
1	1.0005	KLOTILDA ROUBÍČKOVÁ	und	latin	—
2	1.0001	GIUSEPPE SIEBZEHNER	und	latin	—
3	1.0000	LULLUTAN WATERFRONT	und	latin	—
4	1.0000	[illegible]	gu	latin	—
5	1.0000	[illegible]	or	latin	—
6	1.0000	[illegible]	or	latin	—
7	1.0000	[illegible]	kn	latin	—
8	1.0000	[illegible]	kn	latin	—
9	1.0000	Антиохия Писидийска	bg	cyrillic	—
10	1.0000	Античная винодельня	und	cyrillic	—
11	1.0000	Ануфрава Янішэўскіх	be	cyrillic	—
12	1.0000	Акадэмія Спрынгуэст	be	cyrillic	—
13	1.0000	Акадэмія мастацтваў	be	cyrillic	—
14	1.0000	Акадэмія мастацтваў	be	cyrillic	—
15	1.0000	κομητεία Άρμστρονγκ	el	greek	—

Table 41: Cross-script neighbours for **Χάνι Δελβινάκι** [und, greek] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Χάνι Δελβινάκι	und	greek	—
1	1.0005	LALA MONTERREY	und	latin	—
2	1.0005	MANN RESIDENCY	und	latin	—
3	1.0001	EYRE MOUNTAINS	und	latin	—
4	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
5	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
6	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
7	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
8	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
9	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
10	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
11	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
12	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
13	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	arz	arabic	—
14	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	und	arabic	—
15	1.0000	⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧⌧⌧⌧	und	arabic	—

Table 42: Cross-script neighbours for **Κορυφή Ηλείας** [el, greek] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Κορυφή Ηλείας	el	greek	—
1	1.0003	KAFUTA TUMBUN	und	latin	—
2	1.0003	KARIOI FOREST	und	latin	—
3	1.0003	KATARI GARDEN	und	latin	—
4	1.0003	KD⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧	und	latin	—
5	1.0003	KD⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧	und	latin	—
6	1.0003	KD⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧	und	latin	—
7	1.0003	NX⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧	und	cjk	—
8	1.0003	NATURA GARDEN	und	latin	—
9	1.0003	NORDBØ NORDRE	nb	latin	—
10	1.0003	MESSAK MELLET	und	latin	—
11	1.0003	MILUTO KUMANO	und	latin	—
12	1.0003	URAMBO MARKET	und	latin	—
13	1.0003	UR⌧⌧⌧⌧ ⌧⌧⌧⌧⌧⌧	und	cjk	—
14	1.0002	EKEOHA MARKET	igl	latin	—
15	1.0002	EMBUNG KEMIRI	und	latin	—

Table 43: Cross-script neighbours for **Λόνγκμπριτζ** [el, greek] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Λόνγκμπριτζ	el	greek	—
1	1.0006	KAKOKEFALOS	und	latin	—
2	1.0006	KARANGAMPEL	und	latin	—
3	1.0006	KAZIKATONTO	und	latin	—
4	1.0006	KCC⊠⊠⊠⊠⊠⊠⊠⊠	und	latin	—
5	1.0006	NTT⊠⊠⊠ ⊠⊠⊠⊠	und	cjk	—
6	1.0006	NIEMENAIKKU	nb	latin	—
7	1.0006	MASI⊠⊠⊠⊠⊠⊠⊠	hy	armenian	—
8	1.0006	UR⊠⊠⊠⊠⊠⊠⊠⊠⊠	und	cjk	—
9	1.0001	EDAKULPALLY	und	latin	—
10	1.0001	EIKJEHAUGEN	nb	latin	—
11	1.0001	EQUICENTRUM	und	latin	—
12	1.0000	⊠⊠⊠⊠ ⊠⊠⊠⊠⊠⊠	und	cjk	—
13	1.0000	⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠	und	cjk	—
14	1.0000	⊠⊠⊠⊠⊠⊠⊠⊠C⊠⊠	und	cjk	—
15	1.0000	⊠⊠⊠⊠⊠⊠⊠⊠D⊠⊠	und	cjk	—

Table 44: Cross-script neighbours for **Καραβούλα** [el, greek] (14/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Καραβούλα	el	greek	—
1	1.0005	Κανάκηδες	el	greek	—
2	1.0005	LAKHNADON	ga	latin	—
3	1.0005	LG⊠⊠⊠⊠⊠⊠⊠	und	cjk	—
4	1.0005	LG⊠⊠⊠⊠⊠⊠⊠	und	latin	—
5	1.0005	LH⊠⊠⊠⊠⊠⊠⊠	und	latin	—
6	1.0005	LH⊠⊠⊠⊠⊠⊠⊠	und	latin	—
7	1.0005	LOMOTHING	und	latin	—
8	1.0005	MOLLECITO	und	latin	—
9	1.0005	MOMA⊠⊠⊠⊠⊠	und	cjk	—
10	1.0005	MONTAÑITA	und	latin	—
11	1.0005	MOROTONGA	und	latin	—
12	1.0005	L'ALEZANE	und	latin	—
13	1.0005	MALDONADO	und	latin	—
14	1.0005	MANAMBOLO	und	latin	—
15	1.0005	MANIRISOA	und	latin	—

Table 45: Cross-script neighbours for **Ζοριάνο** [el, greek] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Ζοριάνο	el	greek	—
1	1.0018	ECUASAL	und	latin	—
2	1.0018	EDESCON	und	latin	—
3	1.0018	ELBISCO	und	latin	—
4	1.0018	ELHNYNA	und	latin	—
5	1.0018	ELIPSON	und	latin	—
6	1.0018	ELISAVA	eu	latin	—
7	1.0018	ELVENES	nb	latin	—
8	1.0018	ELYSIUM	und	latin	—
9	1.0018	EMASTRU	und	latin	—
10	1.0018	EMBALSE	und	latin	—
11	1.0018	EMBRAPA	und	latin	—
12	1.0018	ENCOMIL	und	latin	—
13	1.0018	ENGEVIX	und	latin	—
14	1.0018	EPITECH	pcd	latin	—
15	1.0018	ERABPRP	und	latin	—

Table 46: Cross-script neighbours for **Χαναάν** [grc, greek] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Χαναάν	grc	greek	—
1	1.0001	KABADI	und	latin	—
2	1.0001	KABETE	und	latin	—
3	1.0001	KABOBI	und	latin	—
4	1.0001	KAHRIF	und	latin	—
5	1.0001	KAKOMA	und	latin	—
6	1.0001	KAMTHE	und	latin	—
7	1.0001	KAROGO	und	latin	—
8	1.0001	KARSEC	gur	latin	—
9	1.0001	KASERE	und	latin	—
10	1.0001	KATECO	gpe	latin	—
11	1.0001	KBS⊠⊠⊠	und	latin	—
12	1.0001	KCC⊠⊠⊠	und	latin	—
13	1.0001	KCHUGR	sl	latin	—
14	1.0001	KEGGER	und	latin	—
15	1.0001	KERNEL	und	latin	—

Table 47: Cross-script neighbours for [illegible] [illegible] [fa, arabic] (0/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	[illegible] [illegible]	fa	arabic	kvh hvrjdh
1	0.9994	[illegible] [illegible]	fa	arabic	dvl ɣʃlɒɣj
2	0.9994	[illegible] [illegible]	fa	arabic	dhʔ sl□mɒn
3	0.9994	[illegible] [illegible]	fa	arabic	dvm vjt͡ʃvsv
4	0.9994	[illegible] [illegible]	fa	arabic	dvn mɒtjɒs
5	0.9994	[illegible] [illegible]	fa	arabic	dvz bjljd͡ʒj
6	0.9994	[illegible] [illegible]	fa	arabic	dvh drhsj
7	0.9994	[illegible] [illegible]	fa	arabic	dær zærgeh
8	0.9994	[illegible] [illegible]	fa	arabic	deh svxteh
9	0.9994	[illegible] [illegible]	fa	arabic	deh ɣ□dræt
10	0.9994	[illegible] [illegible]	fa	arabic	d□m dære□h
11	0.9994	[illegible] [illegible]	fa	arabic	djr ʃrbvrn
12	0.9994	[illegible] [illegible]	fa	arabic	djr ɒlɣrbj
13	0.9994	[illegible] [illegible]	fa	arabic	frt vɒʃɒkj
14	0.9994	[illegible] [illegible]	fa	arabic	frn ɒjkjrz
15	0.9994	[illegible] [illegible]	fa	arabic	fvr vjnts□

Table 48: Cross-script neighbours for [illegible] [illegible] [illegible] [ary, arabic] (6/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	[illegible] [illegible] [illegible]	ary	arabic	—
1	1.0000	[illegible] [illegible] [illegible]	azb	arabic	—
2	1.0000	[illegible] [illegible] [illegible]	azb	arabic	—
3	1.0000	[illegible] [illegible] [illegible]	lrc	arabic	—
4	1.0000	[illegible] [illegible] [illegible]	azb	arabic	—
5	1.0000	[illegible] [illegible] [illegible]	azb	arabic	—
6	1.0000	[illegible] [illegible] [illegible]	azb	arabic	—
7	1.0000	Женщина и суворовец	und	cyrillic	—
8	1.0000	Παναγία η Χρυσοπηγή	el	greek	—
9	1.0000	[illegible] [illegible] [illegible]	azb	arabic	—
10	1.0000	[illegible] [illegible] [illegible]	azb	arabic	—
11	1.0000	הביגדסגעד ö רונמאַר	yi	hebrew	—
12	1.0000	Джерело у Криничної	und	cyrillic	—
13	1.0000	Уиндзор и Мейдънхед	bg	cyrillic	—
14	1.0000	[illegible] [illegible] [illegible]	azb	arabic	—
15	1.0000	Девочка с косичками	und	cyrillic	—

Table 49: Cross-script neighbours for ☒☒☒☒☒☒☒ [ar, arabic] (0/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	☒☒☒☒☒☒☒	ar	arabic	als□u:i:ħ
1	0.9999	☒☒☒☒☒☒☒	ar	arabic	bru:ɣri:s
2	0.9999	☒☒☒☒☒☒☒	ar	arabic	bru:nzu:k
3	0.9999	☒☒☒☒☒☒☒	ar	arabic	bru:nsu:n
4	0.9999	☒☒☒☒☒☒☒	ar	arabic	bru:nfi:l
5	0.9999	☒☒☒☒☒☒☒	ar	arabic	bru:ni:nɣ
6	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:sti:□
7	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:su:i:r
8	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:sˤi:□□
9	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:ɣa:nu:
10	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:ɣi:nz
11	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:fa:rd
12	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:fi:li:
13	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:lku:m
14	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:mru:z
15	0.9999	☒☒☒☒☒☒☒	ar	arabic	bri:mu:rs

Table 50: Cross-script neighbours for ☒☒☒☒☒☒☒ ☒☒☒☒☒☒ [glk, arabic] (12/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	☒☒☒☒☒☒☒ ☒☒☒☒☒☒	glk	arabic	—
1	1.0012	KAMPUNG AINDUA	und	latin	—
2	1.0012	NUESTRA SEÑORA	und	latin	—
3	1.0006	KAMPUNG PAROTE	und	latin	—
4	1.0006	KIRAOLI TEHSIL	und	latin	—
5	1.0006	KVAMMEN SØNDRE	nb	latin	—
6	1.0005	LAGUNAS TAMBOR	und	latin	—
7	1.0005	MANGATU FOREST	ceb	latin	—
8	1.0005	MANGATU FOREST	und	latin	—
9	1.0005	MARIGOT WODORA	und	latin	—
10	1.0000	ACARSOY SİTESİ	und	latin	—
11	1.0000	☒☒☒☒☒☒☒ ☒☒☒☒☒☒	und	latin	—
12	1.0000	☒☒☒☒☒☒☒ ☒☒☒☒☒☒	und	cjk	—
13	1.0000	☒☒☒☒☒☒☒ ☒☒☒☒☒☒	azb	arabic	—
14	1.0000	☒☒☒☒☒☒☒ ☒☒☒☒☒☒	azb	arabic	—
15	1.0000	☒☒☒☒☒☒☒ ☒☒☒☒☒☒	mzn	arabic	—

Table 51: Cross-script neighbours for **Id Ouanir [illegible] [illegible]** [und, latin] (9/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	Id Ouanir [illegible] [illegible]	und	latin	—
1	0.9934	[illegible] 2025 [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]	ks	arabic	—
2	0.9924	К.В. Иванов ячĕллĕ Чăваш патшалăх академи драма театрĕ	cv	cyrillic	—
3	0.9913	Hotel DOMUS SELECTA PALACIO GUEVARA	und	latin	—
4	0.9912	O.N.E.E [illegible] [illegible] [illegible] [illegible] [illegible]	und	arabic	—
5	0.9908	А. С. Пушкин исемендәге китапхана- уҡыу бүлмәһе	ba	cyrillic	—
6	0.9908	אפריקה דרום של החוקתי המשפט בית	he	hebrew	—
7	0.9902	[illegible] [illegible]. [illegible]. [illegible] [illegible] [illegible] [illegible]	gu	latin	—
8	0.9901	התיכון במזרח ויישומי ניסויי למדע האור סינכרוטרון ססמי-	he	hebrew	—
9	0.9898	טנריף של הקונגרסים ומרכז בין-לאומי סחר יריד	he	hebrew	—
10	0.9896	GALPÃO 03 DO COMPLEXO DO HANGAR NAUTICO DA UFRJ	und	latin	—
11	0.9892	אביב בתל טייפה של והתרבותית הכלכלית הנציגות משרד	he	hebrew	—
12	0.9892	[illegible]- [illegible] [illegible]- [illegible] [illegible]	kn	latin	—
13	0.9889	P48: AKO AMAZAKOUÉ ASAN BÉTÉ BI DABÉMA KOTIBÉ SAMBA	und	latin	—
14	0.9888	Hotel COSTA DOIRO AMBIANCE VILLAGE	und	latin	—
15	0.9888	טכנית והכשרה לקריירה וסטינגהאוס ג’ורג’ תיכון	he	hebrew	—

Table 52: Cross-script neighbours for [illegible] [glk, arabic] (15/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	—	[illegible]	glk	arabic	—
1	1.0009	KAIST[illegible]	und	latin	—
2	1.0009	KAKKATHODU	und	latin	—
3	1.0009	KANANGOROK	und	latin	—
4	1.0009	KEELAPOODI	ast	latin	—
5	1.0009	KEELAPOODI	lld	latin	—
6	1.0009	NSK[illegible]	und	latin	—
7	1.0009	NTT[illegible]	und	latin	—
8	1.0009	NTT[illegible]	und	cjk	—
9	1.0009	Μέτσινγκεν	el	greek	—
10	1.0009	NANDOBIGUE	und	latin	—
11	1.0009	NARANJALES	und	latin	—
12	1.0009	NECROPOLIS	und	latin	—
13	1.0009	NIGRILLANI	und	latin	—
14	1.0009	NISSHA[illegible]	und	latin	—
15	1.0009	NKOLABANDA	und	latin	—

Table 53: Cross-script neighbours for ⊠⊠⊠⊠ [ar, arabic] (0/15 cross-script)

Rank	Score	Name	Lang	Script	IPA
Q	–	⊠⊠⊠⊠	ar	arabic	msaːr
1	1.0020	⊠⊠⊠⊠	ar	arabic	uːrʃt
2	1.0020	⊠⊠⊠⊠	ar	arabic	uːruː□
3	1.0020	⊠⊠⊠⊠	ar	arabic	uːriː□
4	1.0020	⊠⊠⊠⊠	ar	arabic	uːr□iː
5	1.0020	⊠⊠⊠⊠	ar	arabic	mqnaː
6	1.0020	⊠⊠⊠⊠	ar	arabic	mqnb
7	1.0020	⊠⊠⊠⊠	ar	arabic	mqnʕ
8	1.0020	⊠⊠⊠⊠	ar	arabic	mhdˤm
9	1.0020	⊠⊠⊠⊠	ar	arabic	mkrm
10	1.0020	⊠⊠⊠⊠	ar	arabic	mqtˤʕ
11	1.0020	⊠⊠⊠⊠	ar	arabic	mnʃuː
12	1.0020	⊠⊠⊠⊠	ar	arabic	mlm□
13	1.0020	⊠⊠⊠⊠	ar	arabic	mmdiː
14	1.0020	⊠⊠⊠⊠	ar	arabic	mnqs
15	1.0020	⊠⊠⊠⊠	ar	arabic	mnqsˤ

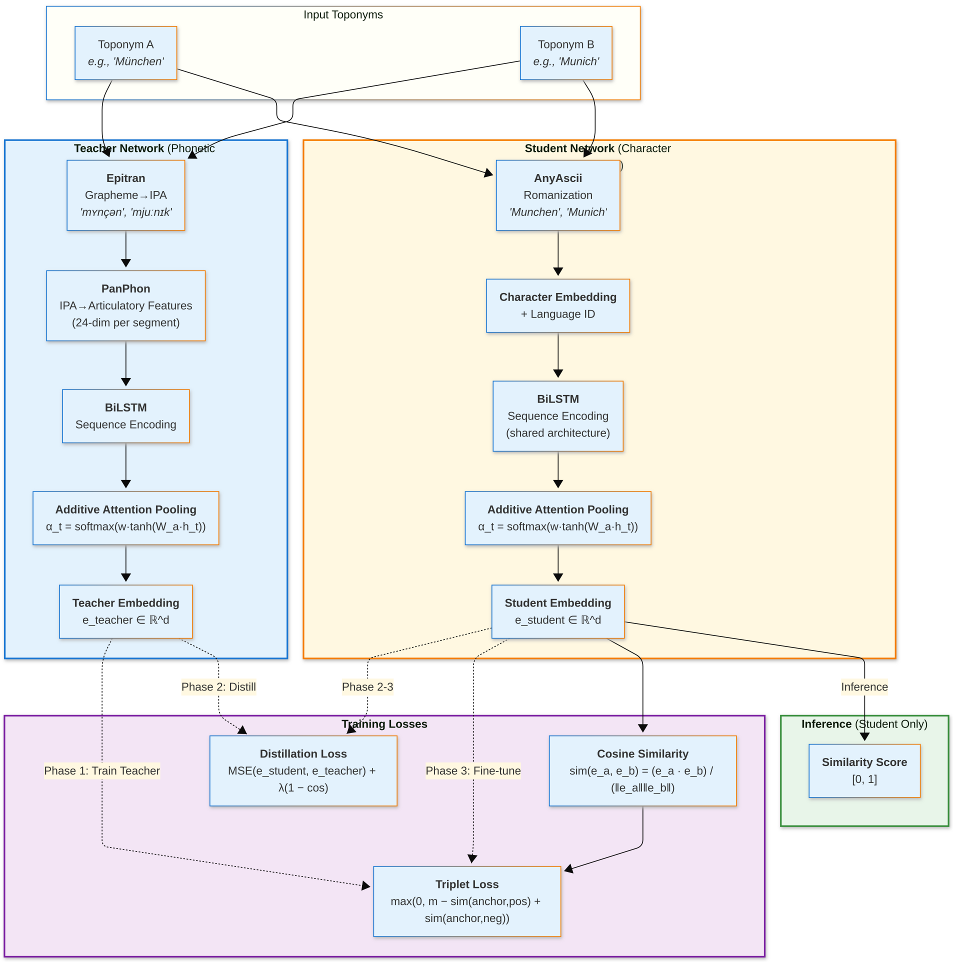

Figure 1: Student-Teacher architecture for phonetic similarity learning. The Teacher network processes IPA articulatory features via Epitran and PanPhon; the Student network processes romanized characters with language embeddings. Both use BiLSTM encoding followed by additive attention pooling. During training, the Student learns to approximate Teacher embeddings through knowledge distillation (Phases 1–3). At inference, only the Student network is required, outputting cosine similarity scores.

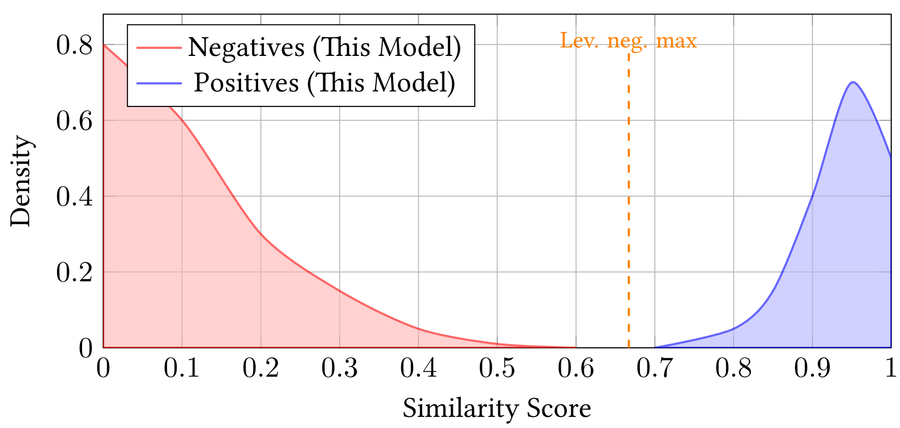

Figure 2: Similarity score distributions for positive and negative pairs. The trained model concentrates negative pairs near zero (95th percentile = 0.485), creating a wide margin for threshold selection. Levenshtein negatives extend to 0.667, reducing the viable operating range.